

Fundamental Inflection Alpha

Four ways to express one company-level investment thesis

Growth acceleration. Improving cash economics. Valuation discipline.

01	Balanced ETFs	15% drawdown criterion
02	Leveraged ETFs	35% proposed drawdown criterion
03	Direct stocks	25% proposed drawdown criterion
04	Leveraged stocks	40% proposed drawdown criterion

Prepared 14 September 2026

The riskier versions reflect the owner's later permission for higher drawdowns. Revised exact limits are proposed settings. These limits are research acceptance criteria, not guarantees against market gaps.

Status: independently unreviewed design. No measured alpha or backtest results. Includes complete entry, exit, allocation, cost and validation rules.

Contents

1. Decision summary	4
2. Economic thesis and boundaries	4
3. Point-in-time information contract	5
4. Stock universe and hard gates	6
5. Valuation definitions and score	7
6. Cash accounting and 12-month bridge	8
7. Catalysts, attention and event risk	9
8. Translating company evidence into ETF evidence	10
9. Shared weighting, volatility and risk mechanics	10
10. Version 1 - Balanced fundamental ETF rotation	12
11. Version 2 - Leveraged fundamental ETF rotation	13
12. Version 3 - Direct-stock fundamental rotation	14
13. Version 4 - Selectively leveraged stock rotation	15
14. Execution, costs, financing and capacity	16
15. Benchmarks and the definition of market alpha	18
16. Stock-first experiment and walk-forward protocol	19
17. Statistical allocation and permanent research ledger	20
18. Acceptance, failure and required report	20
19. Implementation sequence and interfaces	22
20. Sources, provenance and current disposition	24

Reading guide

Sections 1-7 define the investment thesis and company evidence. Sections 8-13 translate it into the four portfolios. Sections 14-19 define execution, costs, validation and the implementation sequence. Sources appear in section 20.

Purpose and status

Prepared 2026-09-14. Status: design candidate, independently unreviewed.

This blueprint turns the owner's small-cap growth, valuation, profitability, cash-flow and catalyst screen into four deterministic investment research strategies. It follows the existing Analyst Revisions, Insider Buying and Short Interest blueprints: stock evidence first, point-in-time aggregation, investable portfolios, costs, independent validation and bounded implementation stages.

The owner initially selected 15% / 25% / 20% / 30% maximum drawdowns, then expressly permitted higher drawdowns for riskier versions, especially those using leverage. The revised proposed limits are **15% / 35% / 25% / 40%** for V1 / V2 / V3 / V4. The later permission is recorded; the revised exact numbers are design assumptions, not a claim that the owner chose those numbers. Numerical settings below are research parameters, not observed market facts, expected returns or approved freezes. There are no backtest results in this document. Outperformance remains a rejection-capable hypothesis. No target prices or current trade list are given.

The editable Markdown is the design source; the companion PDF is its reading edition. Any future conflict must be corrected before adopting a specification. Neither artifact is an executable authorization or an independently accepted research contract.

1. Decision summary

Version	Investable expression	Maximum equity gross / NAV	Volatility target	Maximum drawdown accepted	Required net economic improvement
V1 - Balanced ETF rotation	Ordinary equity ETFs plus cash	1.00x	10% annualized	15%	SPY CAGR +3 percentage points
V2 - Leveraged ETF rotation	The same ETF selection, portfolio margin borrowing	1.75x	18% annualized	35%	Both SSO and QLD CAGR +3 points
V3 - Direct-stock rotation	Long stocks, sales and replacement, plus cash	1.00x	14% annualized	25%	Both V1 and V2 CAGR +2 points
V4 - Leveraged stock rotation	V3 selection with conditional borrowing	1.75x	20% annualized	40%	V3 CAGR +2 points

CAGR means compound annual growth rate, after the complete cost model. A percentage-point difference is not a percentage uplift. Gross exposure measures the value of risky positions relative to net asset value (NAV), after borrowing. The volatility targets scale position size; they do not promise that realized volatility or losses stay within those values.

The numerical economic hurdles give "significantly beat" a concrete meaning. Statistical evidence and risk limits must also pass; beating a benchmark by a small noisy point estimate is insufficient. V3 must beat V2 even if V2 fails its own benchmark goals. V4 must establish a benefit after financing costs.

Design judgment: V3 is the clearest expression of the original stock thesis. V1 offers diversification but may dilute the signal excessively. V2 has the most demanding benchmark/risk combination. V4 is an extension to test only after the unlevered stock mechanism works. These are judgments about testability, not estimates of the probability of success.

2. Economic thesis and boundaries

The proposed edge is delayed recognition of an operating inflection: revenue accelerates, gross margins and cash conversion improve, and financing risk falls before valuation fully reflects the change. A dated commercial or regulatory milestone can make that change visible. The screen also tries to avoid paying an unlimited price for growth.

This mechanism can fail because the information is already priced, consensus is optimistic, reported improvements are accounting effects, or returns are ordinary small-cap, momentum or sector exposure. The research must distinguish those explanations. A company capable of a large price move is not necessarily an attractive portfolio holding at its present price.

The original request for possible 3x-10x stocks motivates searching for asymmetric opportunities. It does not become a portfolio return forecast or a hard signal trained on rare historical winners. Prior discussion of ARQT, ETON, PAY, DLO and AEHR is discovery knowledge, not a frozen security list or untouched validation evidence.

Canonical V3/V4 "buying and selling" means entering long positions and later selling, reducing or replacing them. Short selling, options, inverse funds and single-stock leveraged ETFs are separate hypotheses. Analyst revisions, insider transactions and short interest are not added to this score; otherwise the new mechanism could not be distinguished from the existing lanes.

3. Point-in-time information contract

Every raw observation carries source, stable issuer/security ID, field and units, economic period, public availability time, retrieval time, original content hash, version, correction link and rights status. Estimates additionally carry contributor count, GAAP/adjusted basis and exact fiscal-period identity. Source selection and history availability remain **data to be confirmed**.

Decision clock: fundamental eligibility is updated at 16:00 New York time each trading day. A public non-price fact is first usable at the next trading session's decision cutoff after its verified publication time. Historical source availability must be demonstrable; today's retrieval of a revised historical table cannot establish that an earlier investor knew it. Unknown publication time means refusal. Observed prices through the decision cutoff may be used for that cutoff. Execute only in the following trading session.

This adds a conservative full decision-cycle delay to filings and announcements. It avoids using an after-hours report at that same day's closing price. Actual ingestion delays, if longer, govern subsequent prospective decisions. Amendments change future decisions only; they do not overwrite the historical state.

Use as-filed quarterly accounts. Derive fourth-quarter flows from annual minus the first three quarters only after the annual filing is available. Restated comparatives may inform later decisions but cannot repair an earlier signal. Normalize share classes, splits, prefunded warrants, ticker reuse, mergers, delistings and fiscal-year changes with a dated security master. Never build a historical universe from companies surviving today.

The SEC permits Form 13F reports after quarter end. Accordingly, use the public filing acceptance time, not the holdings date, for institutional information. Late reports, amendments and confidential holdings become usable only when public. [S1]

Freshness limits proposed here: latest quarterly statement period-end no more than 150 calendar days old; consensus snapshot no more than 30 days old; contributing estimate updates no more than 120 days old; catalyst evidence reconfirmed within 90 days; ETF portfolio-holdings publication no more than five sessions old. Institutional comparisons instead use the latest two public quarterly cohorts, with the newer holdings period-end at most 150 calendar days old. Required data outside its limit is unavailable, not silently carried.

4. Stock universe and hard gates

Start from point-in-time US-primary-listed operating-company common equity. Keep historical listings and delistings. This first accounting model excludes banks, insurers, REITs, funds and premerger SPACs, whose margin/cash measures are not comparable. ADRs and foreign-primary listings require a separately defined reporting/identity model. These are disclosed scope restrictions.

Additional investability restrictions: at least 252 listed sessions, latest price at least \$5, and 20-session mean daily dollar turnover at least \$5 million. Use actual traded dollar volume when available; otherwise label close times share volume as an estimate and preapprove that convention before research. Every required gate below must pass. Alternatives within a gate are OR branches; all gates together are AND conditions.

Gate	Canonical proposed rule
Size	Economic equity capitalization \$200 million-\$5 billion, aggregating equivalent common classes and documented prefunded warrants. Use contemporaneous price and latest publicly known share count. Ambiguous economic dilution refuses entry.
Valuation	At least one valid branch in section 5 passes. Missing branches cannot qualify a stock.
Profitability	Three successive quarterly ROE observations improve strictly, with quarterly common net income also improving; OR current negative quarterly ROE plus the documented all-in FCF crossover path in section 6.
Growth	Three-year actual revenue CAGR at least 20%, and FY1 and FY2 consensus revenue growth each at least 15%; OR the quantified order/capacity acceleration branch below.
Gross margin	Latest GAAP quarterly gross margin at least 30% and higher than the year-earlier quarter; OR three consecutive quarterly increases totaling at least one percentage point. All denominators must be positive revenue.
Cash generation	Trailing all-in FCF positive and above its previous four-quarter value; OR the documented crossover bridge below.
Runway	At least 12 months of operating cash support after unavoidable obligations, under the prescribed stressed monthly cash bridge.
Catalyst	At least one sourced, time-bounded milestone within the next 365 days, meeting section 7.
Attention	Matched institutional ownership increases or active research coverage increases, meeting section 7.

Quarterly ROE = quarterly net income attributable to common shareholders divided by average opening/closing common book equity. Both equity balances must be positive. Two consecutive improvements require $Q-2 < Q-1 < Q$; this is not two observations. Net-income improvement prevents a shrinking denominator alone from manufacturing a turnaround. Negative equity makes this ROE branch unavailable. The cash-path branch can still be evaluated when ROE is defined and negative; negative equity is not treated as a positive ROE opportunity.

The original phrase "forward ROE turns positive and grows more than 50%" is not well-defined as ordinary percentage growth from a negative base. That standalone branch is **inactive pending a precise owner-approved definition**. Do not quietly divide by the absolute negative base. TTM ROE is a diagnostic, not a replacement for quarterly ROE in the canonical strategy. These choices are stricter than the earlier hand-screen convention and can exclude ARQT.

FY1 is the fiscal year containing the decision date; FY2 and FY3 are its next two fiscal years. A just-ended year still awaiting actual results uses its dated consensus for growth comparisons and is marked estimated. Require at least

three independent contributors on a consistent basis for each required consensus quantity. An IPO without the required three-year history cannot pass that branch merely through annualization.

Order/capacity alternative: a dated issuer filing/release must identify delivery or commissioning within 12 months and a numeric revenue bridge. Use contracted delivery values or explicit issuer revenue guidance supported by capacity and utilization evidence; do not equate installed capacity with sales. The bridge must imply next-12-month revenue growth at least 15% and at least five percentage points above current TTM revenue growth. Cancellable orders, unknown conversion timing and unnamed aspirational customers cannot establish the bridge. Every conversion assumption is stored, and unsupported items are zero, not an analyst's discretionary probability.

5. Valuation definitions and score

Forward PEG branch:

$$\text{EPS_growth} = (\text{EPS_FY3} / \text{EPS_FY1})^{(1/2)} - 1$$

$$\text{PEG} = (\text{price} / \text{EPS_FY1}) / (100 * \text{EPS_growth})$$

Require both EPS endpoints and EPS growth to be positive. Pass if $\text{PEG} \leq 1.5$. Use diluted per-share consensus on one explicitly identified GAAP or adjusted basis, with the same basis for both endpoints. Record stock compensation and expected dilution separately. Very small positive EPS can produce misleadingly low PEG; flag FY1 net margin below 2% and do not interpret the ratio as proof of cheapness. A vendor's differently defined PEG cannot be substituted.

Peer branch: EV/FY1 sales or equity-value/FY1 sales below the contemporaneous median of at least eight other comparable issuers in the frozen industry classification. Match accounting basis, operating model and positive sales; include qualifying and nonqualifying peers. EV adds debt, preferred equity and noncontrolling interest, then subtracts unrestricted cash/liquid investments. Negative or ambiguous EV makes EV/Sales unavailable; P/S may remain valid.

Own-history branch: current EV/FY1 sales or P/FY1 sales below its median over the preceding 60 month-end point-in-time observations, with at least 36 valid observations. Exclude the current observation from the reference. A current, restated historical ratio chart cannot satisfy this branch.

For each valid valuation branch define a bounded headroom score:

$$v_PEG = \text{clip}(1 - \text{PEG} / 1.5, 0, 1)$$

$$v_peer = \text{clip}(1 - \text{current_multiple} / \text{peer_median}, 0, 1)$$

$$v_history = \text{clip}(1 - \text{current_multiple} / \text{own_median}, 0, 1)$$

The valuation component V is the maximum valid branch score; this OR treatment is fixed in advance. Invalid branches are omitted rather than assigned zero. A passing boundary PEG of exactly 1.5 can qualify while contributing zero headroom. Multiple sales-ratio branches must all follow their frozen definitions.

Shared quality score Q, between 0 and 1:

$$Q = 0.30 * G + 0.25 * P + 0.20 * V + 0.15 * M + 0.10 * A$$

Define $\text{clip}(x, 0, 1)$ as truncating values below 0 or above 1. Use fixed bounded components rather than fitting score weights to realized winners:

- $G = \text{clip}((g - 0.15)/0.35, 0, 1)$. For the consensus growth branch, g is the smaller of FY1 and FY2 growth. For the order branch, g is the supported next-12-month growth. If both qualify, use the smaller valid g.
- $P = \text{clip}((\text{TTM all-in FCF margin} - \text{prior TTM all-in FCF margin})/0.10, 0, 1)$. Margin denominators are revenue for their respective periods. A forward crossover can qualify without receiving credit for improvement not yet seen.

- $M = \text{clip}((\text{latest gross margin} - \text{year-earlier gross margin})/0.10, 0, 1)$. A sequential-improvement qualifier can receive zero if its year-over-year change is negative. Required historical inputs must still exist.
- A is the larger valid attention score defined in section 7, bounded $[0,1]$.

A missing required score input refuses entry; it is not neutral-filled. Rank descending Q with stable security ID as final tie-break. No discretionary "3x probability" or language-model sentiment is part of the calculation.

6. Cash accounting and 12-month bridge

Keep two measures to make acquisition-heavy growth visible:

```
FCF_standard = operating_cash_flow - cash_PP&E - capitalized_software
```

```
FCF_all_in = FCF_standard - cash_product_IP_license_intangible_acquisitions
```

Do not subtract an item twice when it was already deducted in operating cash flow or capital expenditures. Retain whole-business acquisition cash separately in the cash bridge, with any operating-business comparability break disclosed. Dividends, repurchases, debt principal and lease principal are financing cash needs, not extra subtractions from FCF. Principal already included elsewhere is reconciled once. Free cash flow is a defined research measure, not a uniform GAAP line item.

For a crossover, the next four quarters' cumulative all-in FCF must be positive and the fourth-quarter all-in FCF run rate positive. Build a source-linked quarterly bridge from revenue, gross margin, cash operating expense, working capital, capex and committed product/license payments. Management's adjusted EBITDA or non-GAAP EPS guidance alone is insufficient. Any unsourced major input makes the bridge unavailable.

For runway, start with unrestricted cash and saleable short-term investments. Deduct known debt maturities, lease principal, binding purchase obligations, committed license/milestone payments and announced mandatory cash uses due within 12 months. Do not count restricted cash, undrawn/uncommitted borrowing, an assumed equity raise or contingent milestone receipts as available cash.

Construct 12 monthly balances. Use contractual monthly payment dates; absent dates, place a known payment at the beginning of its stated quarter, or month one if only a year is given. Spread quarterly operating flows evenly only when no better monthly schedule exists, and label that assumption.

Use one dated cash ledger: starting cash plus operating/investing flows less financing uses. Each payment ID is charged exactly once. License payments in projected all-in FCF are classified and scheduled as obligations in that ledger, not subtracted a second time. Reconcile the FCF bridge and obligation schedule to the same monthly cash balances before accepting runway.

The stressed operating case reduces the bridge's revenue by 20%, reduces gross margin by five percentage points, holds fixed cash operating costs unchanged and leaves all committed payments intact. Historical cash-burn runway using the worse of recent four-quarter average burn and latest-quarter annualized burn is an additional check, not a substitute for obligations.

Pass runway only if every stressed month-end balance is positive and the final balance covers one further month of stressed cash operating outflows. Do not call a company funded for a year because a simple cash/burn ratio ignores an acquisition payment due next month. These stress magnitudes are proposed screen settings; they are not issuer forecasts or observed cash needs.

7. Catalysts, attention and event risk

Accepted catalyst types are a quantified commercial launch/ramp, signed-order delivery, supported capacity commissioning, a specified regulatory decision, an industry-policy implementation tied to the issuer's economics, or a binding transaction milestone. Store event ID, primary source, publication time, earliest/latest expected date, business linkage, numeric milestone where applicable, cancellation status and binary-event flag.

A general addressable-market story, favorable industry headline, rumor or undated "large customer" does not qualify. If a milestone window expires without updated public evidence, eligibility fails. An announced delay is a new version; it cannot be backdated to preserve an earlier signal.

Institutional route: use the same manager IDs across two adjacent reported quarters. Adjust share counts for splits and issuer changes; exclude options, duplicate manager filings and unresolved amendments. Matched managers must represent at least 60% of the prior quarter's disclosed institutional common holdings for that issuer, with at least three distinct managers increasing shares and positive aggregate matched-manager share growth.

```
A_institutional = clip(matched_share_growth / 0.20, 0, 1)
```

This measures change within a disclosed cohort, not total current ownership or proof of fresh buying today. Do not infer motives or distinguish active from passive flows unless the frozen source permits it. [S1]

Research route: active independent coverage increases by at least two firms compared with the point-in-time count 90 days earlier. "Active" means an attributable published estimate/research update within 120 days. Managers, analysts and brokerage firms are not interchangeable units.

```
A_research = clip(net_new_firms / 5, 0, 1)
```

Set A to the larger valid attention score. At least one attention route must pass; missing alternative coverage is reported. Rumored institutional meetings or unsourced claims of increasing interest do not count.

For direct stocks, define the event window as five trading sessions before through one session after a binary clinical/regulatory decision's announced window. If only a quarter is known, use its entire quarter as the event window. During this window cap each affected issuer at 2% NAV and their aggregate at 10% NAV. V4 cannot finance that sleeve. A surprise binary event remains a gap risk. ETFs apply the same 10% NAV aggregate lookthrough binary-event cap.

8. Translating company evidence into ETF evidence

The \$200 million-\$5 billion requirement applies to companies, not ETF assets. The ETF itself need not have that market capitalization. Do not screen using a fund's blended ROE, a headline P/E or the score of its single best holding.

Let h_{ei} be company i 's disclosed fraction of ETF e 's **total NAV**, P_i its hard-gate pass indicator and Q_i its shared quality score:

$$C_e = \sum(h_{ei} * P_i)$$

$$S_e = \sum(h_{ei} * P_i * Q_i)$$

$$N_{eff_e} = C_e^2 / \sum((h_{ei} * P_i)^2)$$

C is qualifying exposure; S is the diluted fund signal; N_{eff} measures the effective number of qualifying issuers. Unqualified or missing holdings, foreign holdings, cash and derivatives remain in the NAV denominator and contribute no positive signal. Do not renormalize a 1% qualifying sleeve to 100%.

Candidate funds are US-listed transparent ordinary long-only equity ETFs with 252 usable sessions. Exclude leverage, inverse exposure, options-overlay funds, ETNs and opaque active products from selection. Require at least 95% of NAV mapped and accounted for, holdings no older than five sessions, at least 20% qualifying NAV, five distinct qualifying issuers and $N_{eff} \geq 5$.

Also require 20-session average daily dollar turnover $\geq \$20$ million and trailing 30-calendar-day median quoted spread ≤ 20 basis points. Historical spread and holdings availability must be verified before empirical use. For covered transparent ETFs, the SEC rule requires disclosure of prior-day holdings before opening; this does not authenticate a historical archive or apply to every ETF structure. [S2]

Rank all individually eligible funds by S ; keep the complete uncapped raw rank list. Ties use lower measured spread, then stable fund ID. Weighted holdings overlap is $\sum_i \min(h_{ei}, h_{fi})$ after issuer aggregation. Greedily accept funds in the incumbent-then-entry order specified for V1, skipping overlap above 60% with any already accepted fund, up to six. There is no preliminary top-six truncation that would erase the incumbent buffer. Use all mapped holdings for overlap, not just qualifiers.

At least two nonduplicate funds are required to initiate an ETF portfolio. Insufficient funds means cash, recorded as infeasibility/underfill. A universe that cannot satisfy the 20% exposure gate does not support the ETF thesis. Dropping that threshold after seeing returns would create a new hypothesis.

9. Shared weighting, volatility and risk mechanics

At each scheduled selection date, retain eligible incumbents within the specified rank buffer before filling vacancies from the entry ranking. Ties use stable IDs. Allocate base weights in proportion to inverse 60-session annualized volatility, using floors of 10% for ETFs and 20% for stocks. Apply a breadth multiplier $\min(1, \text{eligible_selected_count} / \text{target_count})$, so a small portfolio is not silently scaled to full investment.

Estimate portfolio volatility using the proposed target composition and 20-, 60- and 252-session sample covariance matrices of synchronized daily total returns. Each uses 50% shrinkage to its diagonal. Use the largest of the three annualized portfolio-volatility estimates. Every held instrument needs all 252 observations; do not drop bad dates or replace missing returns with zero. A missing required risk input prevents additions and leverage.

For V1-V3, let u be the normalized unit-gross risky basket, σ_u its forecast volatility, B the breadth multiplier and L the currently allowed gross ceiling:

$$\text{equity_exposure} = \min(B * L, \text{volatility_target} / \sigma_u)$$

Target risky weights initially equal equity_exposure times u. For example, 30% unit-basket volatility and an 18% target allow at most 0.60x exposure, regardless of a 1.75x ceiling. Do not multiply that volatility ratio by the ceiling again. V4's restricted incremental sleeve uses section 13 instead. Drawdown, trend, concentration, liquidity and margin constraints can reduce exposure further. A zero/invalid volatility estimate refuses additions. No expected-return optimizer fits weights to backtest outcomes.

For concentration, first cap individual weights, then proportionally scale each breached sector, then the binary-event sleeve, then the whole portfolio for any remaining constraint. Do not redistribute clipped weight. Inverse volatility normalization precedes these cuts; all released capital stays cash or reduces borrowing. Cash interest and financing are accounted separately.

Drawdown = $1 - \text{current net liquidation NAV} / \text{historical maximum net liquidation NAV}$. Mark at least daily and after simulated fills; include costs, interest, dividends, halted holdings and terminal losses. Never reset the high-water mark after a loss or at a new calendar year. A fresh research epoch does not erase the prior epoch's failed evidence.

Risk constraints operate daily at the cutoff; reductions execute at the next tradable opportunity. Their effect is not retroactive to a gap already suffered. For a reduction state below the permanent-halt threshold, allow a one-step increase only at month-end, after drawdown is at least two percentage points below the threshold that entered that state and trend/risk checks pass. Crossing a deeper threshold immediately replaces the current state.

At the final reduction threshold in the version tables, liquidate risk when executable and keep the frozen run in cash to its end. There is no automatic high-water recovery rule for an all-cash portfolio. A new restart design is a new hypothesis. A halt remains part of the result even if later markets rally.

10. Version 1 - Balanced fundamental ETF rotation

Purpose: express the stock signal through diversified funds, seeking meaningful net SPY outperformance within the owner's 15% drawdown criterion.

- Select monthly at the last trading session's cutoff; trade next session.
- Target six funds. New entries rank in the top six; eligible incumbents can remain through rank nine, subject to the overlap rule and concentration caps. Apply the overlap rule after ordering retained incumbents by current score, then fill from entry ranks. The six-slot cap remains binding.
- Require fund price above its 100-session simple moving average at selection. A daily close below that average triggers exit at the next opportunity.
- Equity gross cap 1.00x and annualized volatility target 10%.
- Maximum 25% NAV per fund, 45% lookthrough per sector, 5% lookthrough per underlying issuer. ETF event exposure follows section 7.
- If SPY closes below its 200-session simple moving average, cap equity gross at 50%. Missing SPY regime data blocks additions and caps equity at 25%.
- Residual unborrowed cash earns the conservative net cash rate in section 14. No substitute momentum ETF fills an unavailable fundamental allocation.

Recompute ETF holdings freshness, company pass flags, qualifying NAV, breadth, lookthrough concentration and event limits daily from usable records. Loss of an ETF hard gate forces its next-executable exit; a portfolio cap breach forces reduction. Falling below two eligible funds exits the remaining ETF portfolio. Rank-only replacement remains monthly, so daily data updates do not authorize discretionary turnover.

Drawdown reached	Maximum equity exposure / action
7.5%	0.50x
10%	0.25x
12%	Exit risk, permanent research halt for this run
More than 15%	Fails the owner's drawdown acceptance condition

How this could work: broad ownership of improving businesses reduces single-company surprises; holding cash when breadth collapses can reduce losses. **What can defeat it:** low qualifying NAV, overlapping sector bets, late fundamentals and trend whipsaws. A strong SPY bull market can make the outperformance objective fail even when risk controls work as intended.

11. Version 2 - Leveraged fundamental ETF rotation

Purpose: test whether selectively financing the same ETF opportunities adds enough net return to beat continuously held 2x market benchmarks, while respecting a proposed 35% maximum-drawdown acceptance criterion.

The primary mechanism is portfolio borrowing against **ordinary selected ETF shares**. This is a leveraged ETF portfolio, not a substitution into an approximately related daily-leveraged sector ticker. Its own brokerage/margin feasibility must later be proven. No broker action is part of this blueprint.

- Use V1's fund selection, rank buffer, holdings tests and monthly schedule.
- Gross ceiling 1.75x; annualized volatility target 18%.
- Financing above 1.00x requires at least three selected, nonduplicate funds; at least three distinct sectors each representing $\geq 10\%$ of the selected unlevered basket; SPY and IWM both above their 200-session averages; and all funds above their 100-session averages.
- Also require at least 20 distinct qualifying issuers across the selected funds after deduplication, current financing inputs and margin headroom.
- If a leverage condition fails, cap gross at 1.00x. V1's lower SPY regime caps still apply. Do not finance cash created by underfill or concentration constraints in order to restore nominal exposure.
- Maximum 35% NAV per ETF, 60% NAV per lookthrough sector, 7.5% per underlying issuer; binary-event exposure still $\leq 10\%$ NAV. Apply these to economic exposure after leverage, not to an imagined unlevered portfolio.
- Equity must be at least 1.5 times modeled maintenance-margin requirements and satisfy initial-margin and buying-power requirements. Any tighter instrument-specific or broker requirement wins.

Drawdown reached	Maximum equity exposure / action
12%	1.00x
20%	0.50x
28%	Exit risk, permanent research halt for this run
More than 35%	Fails the proposed drawdown acceptance condition

The primary challenge benchmarks are **SSO and QLD separately**. Their issuer objectives are twice the daily S&P 500 and Nasdaq-100 returns, respectively; they are not twice those indices' annual returns. [S3, S4, S5]

Compare additionally against SPY and QQQ scaled to V2's own lagged gross exposure and financing convention. That diagnostic distinguishes selected ETF performance from merely changing market exposure. It cannot replace the harder SSO/QLD goals. A 1.75x ceiling may lose to 2x buy-and-hold in a sustained rise; this is a genuine hurdle, not a reason to change benchmarks later.

Actual leveraged-ETF wrappers are not mixed into this canonical experiment. Using them would require exact index mapping, their own point-in-time availability and daily compounding, and a separately registered execution variant. A similarly named sector ETF is not an exact substitute.

Largest failure channels: weak ETF signal amplified by borrowing; trend whipsaw plus financing drag; and sudden margin or liquidity changes. A margin firm may impose higher requirements and liquidate positions to meet a deficiency. The research model must not assume a warning arrives first. [S6]

12. Version 3 - Direct-stock fundamental rotation

Purpose: capture more of the company-level signal and reduce ETF dilution, with enough diversification to seek net superiority over V1 and V2 within the revised proposed 25% maximum-drawdown condition.

- Weekly selection at the last trading session's cutoff; trade next session.
- Target 18 issuers. New entries from ranks 1-18; eligible incumbents can remain through rank 25. Retain incumbents ordered by score, then fill vacancies from entry ranks. If more incumbents qualify than slots, retain the highest scores. Score and hard eligibility are distinct.
- Gross ceiling 1.00x; volatility target 14%; issuer cap 8% NAV; sector cap 25% NAV. Breadth scaling and residual cash apply even if only one stock passes.
- Entry requires price above its 100-session average. Two consecutive daily closes below that average trigger exit. A five-session earnings blackout before and one session after known earnings blocks new entries, while existing holdings remain subject to risk limits and event disclosures.
- If SPY is below its 200-session average, cap equity exposure at 50%; if that regime input is missing, cap at 25% and block additions.
- Recompute fundamental gates daily when new information becomes usable. A failed gate, cash-runway deterioration, missed catalyst, identity failure or stale required input triggers removal at the next executable session.
- Rank-only replacement is weekly. No fixed profit target, automatic sale at 3x, loss averaging or discretionary "story still good" override is used. This lets improving businesses remain held while selling broken conditions.

Drawdown reached	Maximum equity exposure / action
10%	0.75x
15%	0.50x
20%	0.25x
22%	Exit risk, permanent research halt for this run
More than 25%	Fails the proposed drawdown acceptance condition

Buying and selling is driven by a change in eligibility, rank, trend or risk, not an obligation to alternate transactions on a schedule. A sale need not be followed by a replacement if no company qualifies. Data failures block new risk but must not prohibit an otherwise supported exposure reduction.

Largest failure channels: accounting or forecast errors; concentrated binary/gap losses; and turnover/impact that consumes the signal. The 25% drawdown requirement may be particularly difficult for \$200 million-\$5 billion companies. The strategy must demonstrate it rather than infer it from 18 names.

13. Version 4 - Selectively leveraged stock rotation

Purpose: test whether financing the V3 mechanism improves after-cost results without exceeding the revised proposed 40% maximum-drawdown criterion.

- Preserve V3's stock eligibility, ranking, target 18 names, weekly schedule, trend exits and binary-event rules. Each version maintains its own path and drawdown state; leverage-induced exits need not produce identical holdings.
- Gross ceiling 1.75x; volatility target 20%; issuer cap 14% NAV; sector cap 40% NAV. Constraints apply after leverage and do not redistribute clipped positions. Gross above 1.00x is an allowance, not a target.
- Leverage requires at least 12 current holdings across four sectors, each sector at least 10% of the unlevered selected basket; SPY and IWM above their 200-session averages; and verified financing and margin inputs.
- A financed issuer must have 20-session average dollar turnover $\geq \$15$ million. Less liquid eligible issuers may be held in the cash-funded sleeve, but receive no incremental leverage allocation. A financed sleeve must contain at least ten eligible issuers; otherwise cap the portfolio at 1.00x.
- No incremental borrowing funds binary-event-window holdings. Total cash-funded binary-event exposure remains $\leq 10\%$ NAV and $\leq 2\%$ per issuer.
- Apply the same 1.5-times-maintenance-equity buffer as V2. Treat unknown or prohibited instrument margin treatment as no leverage; do not substitute general regulatory minimums for actual brokerage requirements.
- If a leverage gate fails, gross $\leq 1.00x$, further reduced by V3's lower SPY regime cap, volatility and current drawdown state.

Drawdown reached	Maximum equity exposure / action
15%	1.00x
25%	0.75x
32%	0.25x
36%	Exit risk, permanent research halt for this run
More than 40%	Fails the proposed drawdown acceptance condition

Before any use, cash-funded and financed sleeve bookkeeping must enforce a single NAV and borrowing balance: it cannot count the same dollar as both cash reserve and margin support. Sell obligations and financing interest take priority over additions. Unknown buying power prevents more borrowing.

Exact incremental allocation: first compute unlevered base weights b using V4's selected names, 20% volatility target, breadth and risk limits with gross temporarily capped at 1.00x. Apply concentration/event/liquidity cuts without redistribution. Let F be the finance-eligible issuers, $B_F = \sum(b_i \text{ for } i \text{ in } F)$, and $a_i = b_i/B_F$ for i in F , zero otherwise. If B_F is zero or any leverage gate fails, incremental size k is zero. Otherwise choose the largest k in $[0, 0.75*B_F]$ for which $w_i = b_i + k*a_i$ satisfies every current gross, drawdown, volatility, concentration, margin and liquidity constraint. Recompute the three covariance forecasts on w itself; its annualized volatility must be $\leq 20\%$. The feasible upper boundary follows the frozen quadratic/linear constraint solver, rounded downward to one basis point of NAV. No leverage capacity released by an excluded issuer is reallocated to other issuers. Post-rounding checks may only reduce k . Event-window and less-liquid issuers therefore receive exactly their base allocation, never a scaled increment.

Largest failure channels: correlated small-cap gaps, forced deleveraging and financing drag. Report a fixed 1.75x V3 control with identical financing to separate selective leverage from constant leverage. This is an attribution diagnostic, not an additional success claim.

14. Execution, costs, financing and capacity

Research reference capital is **\$1 million per version**, with identical start date and cash. This is a modeling amount, not the owner's account value. Also report fixed capacity diagnostics at \$100,000, \$5 million and \$10 million; those cannot replace the reference-capital primary result.

Signals determined at cutoff t execute next session during 09:45-10:15 New York time against time-stamped prices/quotes. Use marketable limit orders, simulated as paying the contemporaneous ask for purchases or receiving the bid for sales, plus the adverse impact model. A limit is at most 50 basis points beyond the arrival mid for ordinary planned orders. Missing intraday data makes this canonical fill model unavailable; a daily-bar approximation is exploratory. If the simulated bid/ask plus adverse impact crosses the limit price, the quantity is unfilled. Observing the session's high or low does not prove fill.

Ordinary execution may use at most 1% of trailing 20-session daily dollar turnover and 5% of observed window volume. Unfilled purchases expire after three sessions; reevaluate signals before each continuation. Planned sales continue while valid and carry actual inventory/risk until filled. Do not mark an exit at a stale closing price during a trading halt.

Risk and margin reductions override ordinary participation limits but receive adverse stressed fills. They cannot assume unlimited liquidity at the last quote. Forced liquidations, exchange halts, bankruptcy payoffs and corporate action proceeds must be modeled and retained on every relevant date.

Proposed one-way transaction cost, in addition to observed execution mid:

```
cost_bps = quoted_half_spread_bps + max(floor_bps, 0.10 * daily_vol_bps * sqrt(order_notional / ADV20))
```

Floor = 2 basis points for ETFs and 10 for stocks; commission assumption = 0.5 basis points on traded notional, plus the effective dated exchange and regulatory fee schedule. These are starting research assumptions, not claims about a current account's fees. One basis point is 0.01%. Calibrate adequacy using separately approved execution data before freezing outcomes; no post-result reduction of costs is allowed.

The cost formula describes slippage relative to mid. Because bid/ask fills already include half-spread, the cash ledger charges only additional impact, commission and fees on top of those fills. It never deducts half-spread again.

Deduct costs from actual traded notional, not NAV turnover rounded in a table. Record spread, impact, commission and other fees separately. Stock and ETF dividends enter on their economically available dates; do not grant early reinvestment or double count them with adjusted returns. Use a validated total-return ledger for research, actual shares/cash for execution simulation.

V2/V4 borrowing accrues over actual calendar days on the borrowed amount using ACT/360 and the previous available SOFR plus a proposed 300-basis-point spread. If the actual supported account schedule is higher, it governs. The SOFR rate and publication lag must be dated, not copied from today's rate. [S7] For dates before a validated SOFR history, the canonical financing series is unavailable; do not silently splice another rate.

Unborrowed cash earns $\max(0, \text{previous available SOFR} - 100 \text{ basis points})$ under this proposed conservative convention. A known lower account credit rate governs. Never earn cash interest on a negative cash balance. Margin interest, leverage rebalancing and collateral liquidations must all enter net NAV. Actual ETF total-return prices already embed fund expenses and internal financing; do not deduct those embedded costs twice for benchmarks.

Stress cases are fixed diagnostics: double spread/impact; financing at SOFR plus 500 basis points; ADV down 50%; margin requirements raised by ten percentage points; a 10% one-day ETF basket gap; and, separately, a 40% loss in the largest direct-stock holding plus a 15% loss in its other sector holdings. Apply each case and their adverse combination to every scheduled post-rebalance portfolio snapshot and report the worst result; do not select a favorable shock date. These shocks are hypothetical, not estimated probabilities or promised worst cases. Report post-shock equity, liquidity, margin deficit and possible cap breach without smoothing.

Use a conservative model baseline of at least 50% initial and 35% maintenance equity per marginable position unless a verified stricter requirement applies. This is a research assumption, not a statement of universal margin rules. The separate 1.5-times-maintenance buffer makes effective allowed leverage lower when requirements rise. Unverifiable historical margin treatment limits the strength of any leveraged feasibility claim. [S6]

Personal taxes are excluded from canonical after-cost returns because account type and tax circumstances are unspecified. Report turnover and realized-gain characteristics; do not label results after-tax. Leveraged versions must retain a strictly positive net CAGR advantage over every required comparator under doubled trading costs and the higher financing spread. Their full +3/+2-point economic hurdles apply to the canonical cost case, not this stress condition.

15. Benchmarks and the definition of market alpha

Claim ID	Required comparison	Minimum net CAGR advantage	Version drawdown ceiling
FIA-E1	V1 minus SPY	+3 percentage points	15%
FIA-E2	V2 minus SSO	+3 points	35%
FIA-E3	V2 minus QLD	+3 points	35%
FIA-E4	V3 minus V1	+2 points	25%
FIA-E5	V3 minus V2	+2 points	25%
FIA-E6	V4 minus V3	+2 points	40%

SPY, SSO and QLD are fixed buy-and-hold total-return challenges. Admit all on the common start date using consistent initial execution costs, hold cash distributions until their modeled reinvestment time, and include terminal liquidation costs. Use actual post-launch price histories with dated corporate actions. Synthetic prelaunch leveraged returns cannot count as confirmation.

Report SPY, QQQ and IWM context, cash, and the matched-exposure SPY/QQQ controls. The larger of SSO/QLD results is not used as a hindsight-switching benchmark; V2 must separately pass both fixed comparisons. Three-times funds are outside the canonical two-times interpretation of the user's unspecified leverage benchmark. Changing to 3x is a specification amendment before testing.

All six comparisons use identical dates and capital conventions. Cash, underfill, late data, unavailable funds and permanent-halt intervals stay in the path. Missing benchmark observations invalidate the common interval; they cannot selectively remove a losing strategy date. Data deficiencies are reported as invalid/inconclusive, not a valid zero return.

Raw superiority is not enough to claim market alpha. For each strategy fit the preregistered monthly excess-return factor regression using market, size, value, profitability, investment, momentum and sector returns. Use all but one sector as deviations from market; fix the omitted sector and a deterministic rank-deficiency refusal rule before outcomes. Report the net intercept and dependence-aware uncertainty. A validated point-in-time factor source and accounting basis are prerequisites, not assumptions about a download. Standard factor definitions are documented by their primary source. [S8]

Also compare the new score with simple growth/value and momentum controls on the same admitted observations. These are falsification diagnostics, not unbudgeted alternative candidates that may rescue a failed strategy. Exposure matching uses only lagged exposures and the frozen finance/cost model.

16. Stock-first experiment and walk-forward protocol

FIA-S0: does Q predict subsequent company returns beyond established controls? Sample monthly, before ETF construction, across the full admitted historical universe. The primary outcome horizon is 126 trading sessions. Retain delisting losses, mergers and unavailable outcomes with explicit dispositions. Do not count 18 stocks each month as 18 independent experiments.

Use trailing five-year training windows and successive one-year development evaluation folds. Fit the return-control model in training only, using log market value, valuation, profitability, prior 12-minus-1-month momentum, volatility, liquidity, earnings surprise and sector indicators. The primary signal statistic is mean date-level Spearman correlation between Q and out-of-sample residual forward returns. Expected sign is positive. Fixed score weights are never optimized by that control fit.

Purge training labels whose 126-session outcomes overlap evaluation; apply an additional 21-session embargo. All stock observations on the same date belong to the same split. Evaluation-fold labels cannot train controls for their own fold. Related issuers share identity groups. Freeze the exact fitting routine, nuisance regularization, minimum cross-section (30 issuers) and missing-control rules before outcome access. Refused dates count toward feasibility, not toward a fabricated information coefficient.

The common company mechanism must pass before claiming either ETF or stock alpha. Failure of ETF feasibility cannot veto a valid V3 mechanism, but a successful ETF market bet cannot rescue a null company signal.

Exact development and confirmation dates are **unassigned** in this design. They depend on verified source history, power and program coordination. Previously seen prices, issuer performance tables and named-stock research are marked discovery exposure. Do not label an arbitrary historical slice "untouched" just because this agent has not backtested it.

The proposed confirmation floor is 60 nonoverlapping monthly portfolio observations, plus a prospective, pre-outcome power study targeting at least 80% power at the allocated threshold for the stated economic effect. This is a floor, not proof of sufficient power. If the required independent history is unavailable, report insufficient evidence and specify the observation period required; do not declare success from a short bullish interval.

Use synchronized circular moving-block bootstrap draws for all portfolio comparisons, 12-month blocks, 50,000 replicates and fixed seed 20260914. For overlapping stock forward labels use six-month date blocks. These are proposed primary settings; synthetic size/power calibration must validate them before freezing. If calibration fails, revise before outcomes and record the revision; do not choose a block length from favorable p-values.

17. Statistical allocation and permanent research ledger

The existing four named strategy lanes already have total two-sided family error allocation 0.05, with permanent 1/80 slots. Their unused slots expire. This new design is not a fifth slot and cannot inherit their data rights, review status, run budgets or final holdout.

Proposed new-family allocation: subject to a later owner-coordinated program decision, allocate a separate FIA family maximum two-sided error budget of 0.05 across seven fixed claims: FIA-S0 and FIA-E1 through FIA-E6. Each receives at most **1/140**, so seven times $1/140 = 0.05$. This would be a new-family guarantee only; it does not preserve a 0.05 guarantee across both the old and new programs. Actual allocation and outcome access remain zero.

For S0 require positive sign and $p \leq 1/140$. For each E claim use the larger of its paired net-outperformance p-value and its strategy's positive factor-alpha p-value; require that conjunction $p \leq 1/140$, positive estimated effects and the economic hurdle. Factor tests are prerequisites, not separate discoveries. V2 needs E2 and E3; V3 needs E4 and E5; V4 needs E6. Every claimed version also needs S0, valid data/execution, its drawdown cap and stress gates.

Paired outperformance tests use monthly log-return differences, with the null centered at zero. Economic CAGR hurdles are additional conditions; statistical significance of a tiny return advantage does not clear them. Report raw p-values and adjusted claim decisions. For simultaneous intervals over the eleven unique component quantities (S0, six paired effects and four factor intercepts), use Bonferroni two-sided component alpha $1/220$, giving joint coverage of at least 95%. Do not call seven claim-level intervals simultaneous over eleven components. This interval construction does not increase the seven-claim testing budget. Never equate a p-value with the probability of future outperformance.

The existing reserved final holdout **2027-09-01 through 2029-08-31** remains unconsumed by this design. A later coordination decision must assign FIA confirmation dates and selection accounting before any outcomes are opened. No results may be borrowed from Analyst Revisions, Insider Buying, Short Interest or Target-Price Revisions to claim this family passed.

Planned budget after such approval: one immutable development outcome batch containing all four versions and fixed diagnostics, then one confirmation batch if its prerequisites pass. Development is exploratory and spends no confirmatory alpha, but is permanently logged as a research look. Every attempt, error, refusal, partial reveal and corrected execution is recorded. A corrected replay never restores an untouched holdout or a consumed look.

Build and fixture-test all four engines before that development batch. Its prebound dependency graph releases S0 first, and releases/interprets portfolio results only if S0 meets the development progression criterion fixed before the run. No intermediate outcome authorizes changing the portfolio rules. If S0 fails, the unreleased portfolio stages remain unopened and the company mechanism closes null. Structural engine construction is not stock evidence.

No grid of score weights, leverage levels or stop thresholds is authorized by four versions. TTM ROE, relaxed ETF breadth, alternative PEG, wrappers, shorts, options and ensembles are named unimplemented alternatives. Testing any of them requires a new recorded hypothesis and budget before its outcomes.

18. Acceptance, failure and required report

A version is accepted as a research result only when all its required claims and operational-feasibility tests pass on the independently reviewed frozen confirmation protocol. Design quality alone is not performance evidence.

Required evidence for every version:

- Complete net equity curve and the same-date benchmark curves; CAGR, maximum drawdown, drawdown duration, annualized volatility, Sharpe/Sortino under a defined cash rate, Calmar and monthly/annual return tables.

- Corrected paired inference and factor-alpha results; beta and sector exposure, turnover, financing, spreads, impact and unrealized/unfilled risk.
- Eligible-universe and fund counts, qualifying NAV, missing-data refusal rates, time in cash, risk halts, concentration and capacity by period.
- Contribution by issuer, sector and year; report exclusion of the five largest positive issuer contributors as a fragility diagnostic. Do not substitute that diagnostic for the complete primary return path.
- Double-cost/higher-financing results and fixed gap/liquidity/margin shocks. Leveraged versions require positive net benchmark advantage under cost stress and no modeled insolvency or unfinanced margin deficit. All versions report whether shocks would exceed the owner's drawdown ceiling; a breach rejects the risk-constrained design under that stated scenario.
- Source-year/period labels on all actual financial data, immutable manifests, code/config identities, experiment IDs, input/output hashes, and every result's valid/null/invalid/inconclusive disposition.

The following are honest failures: no qualifying stocks or ETFs; insufficient independent dates; no point-in-time consensus/catalyst archive; no incremental stock signal; net underperformance; risk-cap violation; or financing/capacity that cannot be supported. A null can close one expression while another has independent merit. None is fixed by deleting losing dates or loosening the screen after observing outcomes.

19. Implementation sequence and interfaces

These are proposed bounded milestones, not instructions to start jobs:

Stage	Deliverable	Exit condition
FIA-0A	Reviewed semantic specification, exact parameters and lineage contracts	Resolve source semantics, ROE branch, global allocation, dates and power protocol; owner freeze
FIA-0B	Source-rights and availability register	Verify vintage accounts, consensus, holdings, catalysts, prices, terminal returns, quotes, factors, financing and margin inputs
FIA-1	Deterministic canonicalization and gate/score engine	Synthetic fixture checks; missing/late/amended/negative-equity cases refuse correctly
FIA-2	Stock-first event-study engine and immutable run ledger	Synthetic validation; outcome release remains closed
FIA-3	V1/V3 portfolio construction and execution simulator	PIT holdings, diversification, underfill and cost identities verified
FIA-4	V2/V4 finance and risk extension	Cash/margin accounting, adverse liquidations and leverage attribution verified
FIA-5	Frozen common-date development and confirmation package	All engines built; independent review, stock-first release graph, allocated looks, power and source gates closed before execution
FIA-6	Observation and later promotion package	Separate decisions for shadow, paper, restricted live and any unattended operation

Use a lane-owned namespace such as `research/fundamental_inflection/` only after implementation is scheduled. Suggested typed records: `FinancialVintage`,

`ConsensusVintage`, `CashBridge`, `CatalystEvent`, `InstitutionalCohort`,

`EligibilityDecision`, `StockScore`, `EtfHoldingsVintage`, `EtfScore`,

`PortfolioTarget`, `RiskState`, `FinancingState`, `ExecutionDisposition` and

`ExperimentManifest`. Each derived record binds its exact parents and as-of time.

No backtest calls live vendor APIs. Immutable permitted custom/precomputed data and a deterministic QuantConnect/LEAN research adapter can be designed after source-processing rights and the engine choice are expressly scheduled. Local fixtures never count as market evidence. Separate research outputs from the existing assistant's human-approved paper execution path.

Core algorithm outline:

```
for each NYSE session t:
    admit only verified facts available by cutoff(t)
    reconcile corporate actions, holdings, cash, interest and pending fills
    evaluate hard company gates; persist pass/refusal and provenance
```

compute company Q only when all required score inputs are valid

daily: recompute ETF hard gates, qualifying NAV, breadth and current caps

on ETF selection dates: compute full ranking and apply overlap selection

on each version's selection date: retain eligible buffer names, fill ranks

compute breadth-scaled inverse-volatility targets

apply trend, volatility, concentration, event, drawdown and margin limits

allow reductions when supported; block additions on missing critical data

schedule next-session executions with costs, limits and unfilled inventory

mark net NAV; retain cash, halted periods and every refusal in the ledger

Dangerous-direction fixture requirements include: filing after cutoff; quarter-end masquerading as availability; amended holdings; reused ticker; negative equity; low-base PEG; double-subtracted licenses; restricted cash; delisted losers; one qualifying ETF holding falsely renormalized; overlapping funds; insufficient breadth; unknown margin; rate publication lag; leverage after a drawdown; stale quotes blocking risk reduction; and a trading halt that prevents a planned exit. Compare independent cash/share/NAV calculations.

20. Sources, provenance and current disposition

Primary public sources below were consulted on **2026-09-14** for mechanics. They do not validate strategy alpha, supply a historical research entitlement or verify current candidate holdings. No source is quoted at length.

- **S1 - SEC, Form 13F FAQ** (current page accessed 2026): reporting schedule, filing requirements and interpretation of disclosed institutional positions.

Open primary source: www.sec.gov

- **S2 - SEC, Exchange-Traded Funds compliance guide** (2019 rule guide, accessed 2026): covered ETF holdings disclosure and excluded structures.

Open primary source: www.sec.gov

- **S3 - ProShares SSO issuer page** (accessed 2026): 2x daily S&P 500 objective.

Open primary source: www.proshares.com

- **S4 - ProShares QLD issuer page** (accessed 2026): 2x daily Nasdaq-100 objective.

Open primary source: www.proshares.com

- **S5 - SEC Investor Bulletin, leveraged and inverse ETFs** (2023): daily reset objectives and divergence over longer holding periods.

Open primary source: www.investor.gov

- **S6 - FINRA, Margin Calls** (accessed 2026): margin changes and liquidation risk.

Open primary source: www.finra.org

- **S7 - Federal Reserve Bank of New York, SOFR** (accessed 2026): secured overnight reference-rate definition and publication framework.

Open primary source: www.newyorkfed.org

- **S8 - Kenneth R. French Data Library** (accessed 2026): primary definitions of standard research factors; historical availability and usable vintages still require a separate data contract.

Open primary source: mba.tuck.dartmouth.edu

Repository design references: the Analyst Revisions V2, Insider Buying and Short Interest PDFs in docs/Strategy Description/; the current Action Plan;

docs/THREE_STRATEGY_PROJECT_DIRECTION.md; and the general review/handoff process. Existing lane contracts and reserved evidence remain unchanged.

Current disposition: **four complete design chapters; no measured returns, no validated alpha, no current selection portfolio, no source acquisition, no outcome run and no implementation/promotion claim.** Numeric design settings, including the revised higher-risk drawdown assumptions, remain proposals for independent review. Any actual market/financial field not verified later must be labeled **data to be confirmed**, never filled with an invented value.